

Planning Fast and Slow: Guided Trajectory Synthesis via Conditional Flow Matching

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I confirm that the project dissertation I am submitting is entirely my own work and that any material used from other sources has been clearly identified and properly acknowledged and referenced. In submitting this final version of my report to the JISC anti-plagiarism software resource, I confirm that my work does not contravene the university regulations on plagiarism as described in the Student Handbook. In so doing I also acknowledge that I may be held to account for any particular instances of uncited work detected by the JISC anti-plagiarism software, or as may be found by the project examiner or project organiser. I also understand that if an allegation of plagiarism is upheld via an Academic Misconduct Hearing, then I may forfeit any credit for this module or a more severe penalty may be agreed.

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WORD COUNT

Number of Pages:

Number of Words:

ABSTRACT

This should provide a summary of what highlights the reader is going to expect to read. The length should be short, 200-300 words. An abstract should include: The significance of the work, which may require a brief overview of the topic area. The contributions that this work is making to the topic area, the key results that are presented Achievements to date, what they were and how they are paving the path for future work.

Note that this is the first thing that the examiner is going to read possibly after taking a glance at the title. They need to be informed about what they are going to expect to find in this report.

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1 INTRODUCTION

Here in the first part of your introduction you need to make an introduction to the topic with one or two paragraphs. State what the topic is, where it is used and what its purpose is. The reader needs to be in no doubt over the topic area you are working in (e.g. communication networks, wireless systems, radio, computer vision, signal processing etc.) and the type of work it involves (e.g. theoretical, experimental, programming or hardware).

1.1 Background and Context

This is where you can briefly expand on the general history of the field of study and allow a general reader to better understand the rationale for your chosen study. Some literature references might be used here – cited appropriately. You may also discuss examples of applications. You should also give a brief idea of where there is requirement for, or motivation for new work in this area which justifies the work you are carrying out in this dissertation. By writing this way, it should then naturally lead onto the next section stating what the objectives are of your work.

1.2 Objectives

The objectives should be clearly listed here to explain what new outputs you plan to achieve in this work. After the abstract, this section is the one next most important thing your examiner will read. They want to know what you plan to do and see if you are on target to achieve it. You will have listed these objectives in your first report that you submitted at the start of your project. Therefore copy and paste them into this section and show them in bullet points each with no more than one sentence like below:

- Objective 1.....
- Objective 2
- Objective 3
- ..

- Objective N

You may find that you need to slightly revise your objectives since writing your first report. This is totally acceptable if you see a need but you are advised to discuss any revisions with your supervisor. Be very clear to understand what is an objective. An objective is a new output providing new knowledge or bringing knowledge together. Therefore statements like “learn about ...”, “perform simulations on...” or “write software code for...” are not objectives, they are tasks that would possibly be required to form an objective. Therefore objectives are not tasks; they are statements of what advances the work aims to or will make.

Having listed your objectives you may wish to summarise below the tasks you have carried out and will carry out to fulfil the objectives, so that for example it gives an idea of what work was involved in the project such as building new hardware, performing simulations or carrying out theoretical analysis.

1.3 Achievements

Summarise what you have achieved in your work, what are the highlights that the reader should look forward to reading about?

1.4 Overview of Dissertation

Briefly overview the contents of what follows in the dissertation explaining what is in each chapter and explain it in a way that show the narrative of the work so that it makes sense to the reader why the dissertation is organised in the way it is.

2 BACKGROUND THEORY AND LITERATURE REVIEW

This chapter is required to include a more deep literature review beyond what you briefly introduced in section 1.1. Furthermore it should include the background theory and concepts to what is critical to your work. It is very important to consider what sections you should form within this chapter and how to structure it. You should send a planned contents of this and other chapters to your supervisor before you write up and ensure they are happy with what you propose to do.

The earlier sections of this chapter should include a literature review of the state of the art in your topic area citing books, publications and possibly some other web sources of work relevant to the topic. They should be written up in a way that they lead the reader to realising where your contributions to the work are useful and filling a gap. Don't just assume the reader will know what you write about, as your second examiner is not considered to be expert in the area and will want to learn from what you write about.

There will be background theory and concepts that are directly relevant to your work. It is important to work out what these are and develop appropriate sections for this purpose. Again the reader may need to learn about these so it is important that you explain them clearly and in the context that is relevant to your work, not just repeating what all the other literature says. You need to show you understand it by explaining it.

You should ensure that at this point in the chapter a paragraph or two is presented to show what content will be in this chapter and this significance in order to indicate where it is going to lead the reader.

2.1 Section Title

Text for section

2.2 Section Title

Continue with as many sections as you need

Note that references work like this [3] [2]

2.3 Summary

Your final section should include a summary of the chapter, which will summarise what has been covered and where there are gaps in knowledge which pave the way into the original contributions you make in the coming chapters.

3 TECHNICAL CHAPTER

The technical body of the report consists of a number of chapters (just one here, but there will usually be more) of work you have carried out to date. Follow a logical structure in how you present your work. This will usually be the phases of the work you carried out which will link loosely to the objectives. Note that these chapters are not meant to be a log book, thus do not write them like a diary of work carried out but as a technical report specifying clearly what has been implemented and presented in a way that someone else could carry out the work based on what they have read.

Again you need to consult with your supervisor on the technical chapters you plan to write by providing a draft contents page. It should show clearly the sections in it and every chapter should have a clear paragraph introducing it and what to expect in it.

3.1 First Section

Structure your text into sections.

3.1.1 First Subsection

If necessary, also use subsections. Subsections are entered using the Heading 3 paragraph style (all these heading styles are self-numbering).

3.1.1.1 First Subsubsection

If you really need subsubsections.

3.1.2 Second Subsection

And, as required, more subsections.

3.2 Second Section

As an example of a figure, consider Figure 3.1 and note how it is referenced here by cross referencing to a caption method so that also you can form a list of figures at the start. You should be familiar with how to insert a caption and refer to it in Word from the training needs analysis form you were given. The caption should include title and essential

information about figure and figures must always be referred to in the text. Ensure you insert figures into appropriate points in the text so they are both near to the text that is referring to the figure but also ensure that they are arranged in a tidy fashion so they do not leave big gaps with no text.

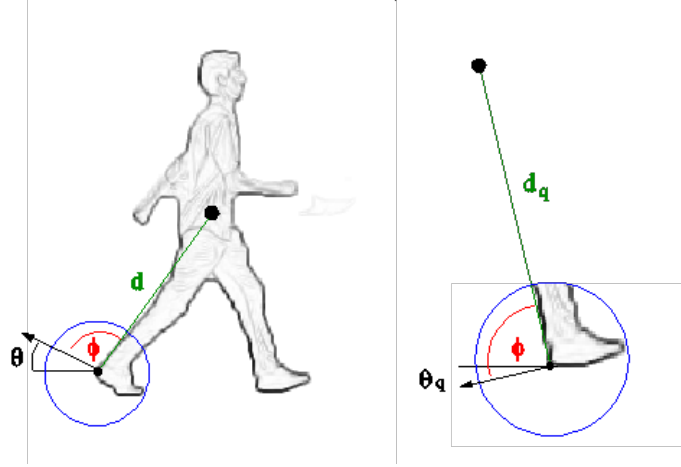


Figure 3.1: Example figure and caption.

Table 3.1: Make sure to set caption to large size.

Test	Test
More	Testing
More	Testing

3.3 Third Section

This sections hows how to do equations and to show that they number correctly as LaTeX can do with the chapter number and equation number to refer to equation in a piece of text:

$$y = x^2 \tag{3.1}$$

but note it is usually after putting in an equation that you may refer back to equation (3.1) later in the document or chapter.

4 TECHNICAL CHAPTER 2

You need to work out how many technical chapters you will require, possibly more than two in this template. Normally your final chapter before the conclusion will include some clear results or proof of concepts from the end goals of your dissertation and it is important to document these well and show the value of what you have achieved. Frequently many dissertations fail to document this information correctly because it is rushed at the end so do ensure you allow time to write it properly.

4.1 Section

Text to insert

4.2 Section

Text to insert – but ensure the last section really does show the value of your work and where you have got to.

5 CONCLUSIONS

Your conclusion should summarise what you have carried out in this work to date, what you have achieved and what you are on course to achieve. Also it should explain what was documented in this dissertation. Note that someone may read this chapter without bothering to read the earlier chapters so there is no problem with repeating what you may have already written earlier. Indeed no new information should be in this part of the conclusion. It should reference back to what is already there.

5.1 Evaluation

Stand back and evaluate what you have achieved and how well you have met the objectives. Evaluate your achievements against your objectives in section 1.2. Demonstrate that you have tackled the project in a professional manner.

5.2 Future Work

Discuss what further work could be carried out from this work and relate it to what has been carried out.

BIBLIOGRAPHY

- [1] ESA. LEON's First Flights. [http://www.esa.int/Our_Activities/Space_Engineering_Technology/LEON_s_first_flights/\(print\)](http://www.esa.int/Our_Activities/Space_Engineering_Technology/LEON_s_first_flights/(print)), 2016. Accessed: 3-12-2016.
- [2] RTEMS C User Guide. RTEMS C User Guide. On-Line Applications Research Corporation, Feb. 2013.
- [3] Wall, J. V. , Shimmins, A. J. , and Bolton, J. G. . The Parkes 2700 MHz Survey (Ninth Part): Supplementary Catalogue for the Declination zone -45° to -65° . Australian Journal of Physics Astrophysical Supplement, 1975.
- [4] Wertz, J. R. and Larson, W. J. . Space Mission Analysis and Design. Space Technology Library. Springer Netherlands, 1999. ISBN 9780792359012.

APPENDIX

You may have one or more appendices containing detail, bulky or reference material that is relevant though supplementary to the main text: perhaps additional specifications, tables or diagrams that would distract the reader if placed in the main part of the dissertation. Make sure that you place appropriate cross-references in the main text to direct the reader to the relevant appendices. Again you should discuss with your supervisor what appendices are appropriate for this work.

Note that you should not include your program listings as an appendix or appendices. You should submit one copy of such bulky text as a separate item, perhaps on a disk.